

datagram passes through routers or nodes, each router or node decreases the TTL field by the default value set by the router or network administrator; minimum value of default being 1. If a router or node receives a datagram with TTL field as 0, and if the datagram does not reach its final destination, the datagram is discarded by that router or node.

- 8-bit protocol field that defines the protocol in which data is encapsulated. For example, when the IP datagram carries a protocol field with 6, it means data is a TCP pack. The standard is already set and available for various encapsulating protocols. For UDP it is 17.

- Header check sum field that is of 16 bits. This field is used to check errors on the header fields of the IP datagram. Checksum is calculated using 1s complement addition rule, which was discussed earlier. It is computed by taking 16-bit words from header fields. The checksum field ensures checking of errors only at link level on header fields.

TCP performs end-to-end error control, which takes care of errors on the data field. That is why at IP level, error checking at the data field is not done. IP header checksum is computed at each router or node, or on hop-to-hop basis, since each router/node changes the TTL field as well during fragmentation of IP headers (like identification, M flag, offset fields, etc).

The next two levels are for addresses. IP header has two address fields: source IP address and destination IP address. These are used to route datagrams. Similarly, addresses are used in postal letters. IP uses four modes of addressing for this purpose. These are known as A-class, B-class, C-class and D-class. The difference in classes is due to the number of bits used for network identification and the number of bits used for host identification.

Different address formats under different classes are illustrated in Fig. 5. IP addresses are of 4 bytes, and these are usually mentioned as four sets of dot-separated octets. For A-class, most significant bit of the first octet is always 0. Therefore the next seven bits are used for network

identification. This means that only 27 to 128 networks can be of A-class. Hosts may be 224.

For B-class address, first two bits of the first octet must be 10. This means the first octet has to be from 128 to 191. 127 is reserved for special purposes. Total networks and hosts under this class are, respectively, 214 and 216.

For C-class address, first three bits of the first octet are 110. This means the first octet will be from 192 to 255. Class-C address has 221 networks and 28 hosts. Minimum hosts under different addressing modes are:

Class A: $2^7 \times 2^{24} = 2^{31}$

Class B: $2^{14} \times 2^{16} = 2^{30}$

Class C: $2^{21} \times 2^8 = 2^{29}$

Address scheme under A-class to C-class is for unicast (one-to-one) communication. Other addressing schemes are for multicast (one-to-many) communication. Class-D and Class-E belong to multicasting.

In D-class, all four octets are used to identify the group of nodes designated to receive a multicast. D-class addresses do not specify the network. These are in the range of 224.0.0.0 to 239.255.255.

E-class addresses range from 240 to 255 in the first octet, and are used for experimentation.

Maximum number of hosts or nodes per network are in A-class, and it is about 2^8 times that of B-class and about 2^{16} times that of C-class. Maximum number of hosts or nodes per network in B-class is about 2^8 times that of C-class. The logical conclusion is that A-class addressing is for larger networks, B-class for medium and C-class for small networks. So, if your network is large, you can apply for A-class address, for medium network, B-class and for small network, C-class address.

An organisation may opt for any one of these classes based on size. Say, an organisation has 2000 hosts under its local network. Thus, it cannot opt for C-class address, because under one network address of C-class, maximum 128 hosts can be addressed. The optimal option is then for B-class, because under a network address of B-class, maximum of 2^{16} hosts can be addressed.



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